CAPSTONE PROJECT

Automated IT Infrastructure Platform

Design, Consolidation, and Extension of an Automated IT Infrastructure Platform for a Small Organization

9	3	3
prior projects consolidated	VMs in the platform	real internal services

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THE PROBLEM

No centralized, automated platform for internal services

Before

Nine disconnected lab projects, manual deployment, no shared documentation, no identity layer, no real services running on top of the infrastructure.

Objectives

- Consolidate 9 prior projects into one platform
- Add centralized identity management
- Deploy 3 real internal services
- Wire everything into monitoring and logging
- Automate with Ansible and CI/CD
- Validate operational readiness, not assume it

SCOPE

What's in, what's a follow-on

In scope

- Docker / Compose on Ubuntu Server
- Nginx reverse proxy, BIND9 DNS
- Prometheus, Grafana, Loki/Promtail
- FreeIPA identity management
- Ansible automation, GitHub Actions CI/CD
- 3 real internal services

Follow-on projects

- SSL/TLS across services
- WireGuard VPN
- Windows Active Directory
- Kubernetes / multi-node / HA

ARCHITECTURE

Three VMs, one internal network

server-vm
10.10.10.10
Nginx, BIND9, monitoring stack, CI/CD runner, Wiki.js, Snipe-IT, Uptime Kuma

client-vm2
10.10.10.20
Promtail · node_exporter · Ansible-managed target

freeipa-vm
10.10.10.30
FreeIPA on AlmaLinux 9.8 · standalone, no integrated DNS

labnet — 10.10.10.0/24 (VirtualBox internal network)

Each VM also carries a NAT adapter for independent internet access during package installs and image pulls.

TECHNOLOGY STACK

Every layer, one stack

Layer	Technology
OS / networking	Ubuntu Server, static IP, SSH, UFW
DNS	BIND9
Containers	Docker, Docker Compose
Reverse proxy	Nginx
Metrics	Prometheus, Grafana, node_exporter, cAdvisor
Logging	Loki, Promtail
Identity	FreeIPA (AlmaLinux 9.8)
Automation	Ansible
CI/CD	GitHub Actions (self-hosted runners)
Services	Wiki.js (Postgres), Snipe-IT (MariaDB), Uptime Kuma

IMPLEMENTATION

Ten phases, all delivered

Phase	Deliverable	Status
Day 1	Infrastructure audit of all 9 prior projects	Complete
Day 2-3	Hardening (secrets, resource allocation)	Complete
Day 4-6	FreeIPA identity management (AlmaLinux 9.8)	Complete
Day 7-9	Wiki.js, Snipe-IT, Uptime Kuma deployed	Complete
Day 10	Monitoring integration	Complete
Day 11	Ansible consolidation, real deploy validated	Complete
Day 12	CI/CD extension (self-hosted runner)	Complete
Drill	Backup/restore validation	Complete
Day 13-14	Architecture docs, troubleshooting guide	Complete
Day 15-16	Report and presentation	This deck

THE BIG LESSON

Host starvation corrupts guest filesystems

Three corruption events on two VMs. When a guest goes without CPU time, the kernel soft-lockup fires and in-flight writes corrupt ext4 at the block level.

Fix: clone-then-resize disks, and never force-close a paused VM.

- DNS fell back to public resolvers: netplan override fixed
- FreeIPA failed on Ubuntu: moved to AlmaLinux 9
- Memory check failed on good VMs: threshold recalibrated
- Playbook used uncommitted files: test from a fresh clone

VALIDATION

Proven, not assumed

CI/CD pipeline

A test push ran the self-hosted runner end to end. Green in Actions.

Ansible validation

--check clean on both hosts. Real run: failed=0.

Backup/restore drill

First restore silently failed ("already exists"). Caught by running it.

Fixed procedure: stop app, drop and recreate DB, restore, restart. Test page gone, wiki back to backup state.

RESULTS

What this platform does today

3	9	2
services deployed by CI/CD	prior projects re-verified	incidents fully recovered

- FreeIPA on AlmaLinux 9: Kerberos login verified
- New containers auto-discovered by monitoring
- Push to the repo triggers a redeploy
- Installation, Administrator and Troubleshooting guides

CONCLUSION

Deployed, monitored, and tested

Most of the extra time went to real troubleshooting: OS mismatches, host resource starvation, DNS, a flawed restore procedure. Every fix was verified against a real failure.

Deferred

- SSL/TLS
- WireGuard VPN
- Windows AD
- Kubernetes / HA

Next

- Drill the MariaDB restore
- Drill VM snapshot restore
- Script the restore end to end

THANK YOU

Questions and live demo

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